

Formalization of Weingarten Calculus

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Chapter 1

Schur Weyl Duality

1.1 Operator Space

Suppose $\mathcal{X} = \mathbb{C}^{[d]}$ and $\mathcal{Y} = \mathbb{C}^{[d]}$ where $d \in \mathbb{N}$. Let the set of linear operators be $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. When $\mathcal{Y} = \mathcal{X}$, let the set of linear operators be $\mathcal{L}(\mathcal{X})$.

We briefly restate some relevant definitions and properties of linear operators that we will use in the subsequent sections.

Remark 1.1.1. For any pair of operators $X, Y \in \mathcal{L}(\mathcal{X})$, the Lie bracket $[X, Y] \in \mathcal{L}(\mathcal{X})$ is defined as

$$[X, Y] = XY - YX, \quad (1.1)$$

such that $[X, Y] = 0$ if and only if X and Y commute.

Remark 1.1.2. Let $\mathcal{A} \subseteq \mathcal{L}(\mathcal{X})$ be a subalgebra of $\mathcal{L}(\mathcal{X})$. The commutant of $\mathcal{A}$ is given by

$$\text{Comm}(\mathcal{A}) = \{Y \in \mathcal{L}(\mathcal{X}) : [X, Y] = 0 \text{ for all } X \in \mathcal{A}\}. \quad (1.2)$$

Definition 1.1.3 (Diagonal operators). An operator $X \in \mathcal{L}(\mathcal{X})$ is a diagonal operator if $X_{a,b} = 0$ for all $a, b \in [d]$ where $a \neq b$. For a given vector $u \in \mathcal{X}$, we can define a diagonal operator $\text{Diag}(u) \in \mathcal{L}(\mathcal{X})$ to denote the diagonal operator whose diagonal entries are given by the entries of u . That is, we have

$$\text{Diag}(u)_{a,b} = \begin{cases} u_a & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases} \quad (1.3)$$

Remark 1.1.4 (Projection operators). Let Π denote the project operator onto a subspace $\mathcal{V}$. Then Π is a linear operator that satisfies the following properties:

- $\Pi^2 = \Pi$, the idempotent property; and
- $\Pi^* = \Pi$, the self-adjoint property.

Let the collection of all projection operators be denoted $\text{Proj}(\mathcal{X})$. There is a uniquely defined projection operator $\Pi \in \text{Proj}(\mathcal{X})$ such that its image $\text{Im}(\Pi) = \mathcal{V}$. When necessary, we denote this projection operator as $\Pi_{\mathcal{V}}$ to indicate the subspace $\mathcal{V}$ onto which it projects.

Remark 1.1.5 (Isometries). An operator $\mathcal{A} \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is an isometry if it satisfies the condition

$$\mathcal{A}^* \mathcal{A} = I_{\mathcal{X}}, \quad (1.4)$$

which is equivalent to preserving the Euclidean norm of vectors in $\mathcal{X}$, i.e., $\|\mathcal{A}|\psi\rangle\| = \| |\psi\rangle \|$ for all $|\psi\rangle \in \mathcal{X}$.

Remark 1.1.6 (Unitary operators). Denote the set of isometries that map $\mathcal{X}$ to itself as the set of unitary operators $U(\mathcal{X})$. Every unitary operator $U \in U(\mathcal{X})$ is necessarily invertible and satisfies the normal condition

$$UU^* = U^*U = I_{\mathcal{X}}. \quad (1.5)$$

1.2 Symmetric subspaces

Definition 1.2.1 (Symmetric subspace of linear operators). The symmetric subspace of linear operators $\mathcal{L}(\mathcal{X})^{\otimes k}$ is the set of all operators $X \in \mathcal{L}(\mathcal{X}^{\otimes k})$ that remain unchanged (invariant) under the permutation W_{π} . That is,

$$\mathcal{L}(\mathcal{X})^{\otimes k} = \{X \in \mathcal{L}(\mathcal{X}^{\otimes k}) : X = W_{\pi} X W_{\pi}^* \text{ for all } \pi \in S_k\}. \quad (1.6)$$

Remark 1.2.2. Let S_k denote the symmetric group. Then given $\pi \in S_k$, the permutation matrix (operator) $V_d(\pi)$ is the unitary matrix that satisfies

$$W_{\pi} |\psi_1\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_{\pi^{-1}(1)}\rangle \otimes \dots \otimes |\psi_{\pi^{-1}(k)}\rangle \quad (1.7)$$

for all $|\psi_1\rangle, \dots, |\psi_k\rangle \in \mathbb{C}^d$.

Definition 1.2.3 (Standard computational basis of linear operators). The standard computational basis of linear operators $\{E_{i,j}\}_{i,j \in [d]}$ is defined as the set of d^2 operators in $\mathcal{L}(\mathcal{X})$ such that

$$E_{i,j} = |i\rangle\langle j| \quad (1.8)$$

for all $i, j \in [d]$, where $\{|i\rangle\}_{i \in [d]}$ is the standard computational basis of $\mathcal{X}$.

Definition 1.2.4 (Operator-vector correspondence). The operator-vector correspondence is a bijective mapping between the set of linear operators $\mathcal{L}(\mathcal{X})$ and the tensor product space $\mathcal{X} \otimes \mathcal{X}$ defined as

$$\text{vec} : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{X} \otimes \mathcal{X}, \quad E_{i,j} \mapsto |i\rangle \otimes |j\rangle \text{ for all } i, j \in [d]. \quad (1.9)$$

One relevant identity we use is that for $A_0, A_1, B \in \mathcal{L}(\mathcal{X})$,

$$(A_0 \otimes A_1) \text{vec}(B) = \text{vec}(A_0 B A_1^T). \quad (1.10)$$

1.3 Permutation Invariance and Properties of Unitarily Invariant Measures

Theorem 1.3.1 (Tensor power span theorem). *Let k be a positive integer, and let $X \in \mathbb{C}^{[d]}$. For any set $\mathcal{A} \subseteq \mathbb{C}$ satisfying $|\mathcal{A}| \geq k + 1$, it holds that*

$$\text{Span}\{v^{\otimes k} : v \in \mathcal{A}^{[d]}\} = \mathcal{X}^{\otimes k}. \quad (1.11)$$

Proposition 1.3.2 (Symmetric subspace and span of tensor product space). *Let $\mathcal{X}$ be a complex Euclidean space and k be a positive integer. Then the following statements are equivalent:*

1. $X \in \mathcal{L}(\mathcal{X})^{\otimes k}$.

2. For $V \in U(\mathcal{X}^{\otimes k} \otimes \mathcal{X}^{\otimes k}, (\mathcal{X} \otimes \mathcal{X})^{\otimes k})$ being the isometry defined by

$$V \text{vec}(Y_1 \otimes \cdots \otimes Y_k) = \text{vec}(Y_1) \otimes \cdots \otimes \text{vec}(Y_k) \quad (1.12)$$

holding for all $Y_1, \dots, Y_k \in \mathcal{L}(\mathcal{X})$, one has that

$$V \text{vec}(X) \in (\mathcal{X} \otimes \mathcal{X})^{\otimes k}. \quad (1.13)$$

3. $X \in \text{Span}\{Y^{\otimes k} : Y \in \mathcal{L}(\mathcal{X})\}$

Theorem 1.3.3 (Equivalence of symmetric subspace and unitary tensor span). *Let $\mathcal{X}$ be a complex Euclidean space and k be a positive integer. Then the symmetric subspace of linear operators is equivalent to the span of the unitary tensor products, i.e.,*

$$\mathcal{L}(\mathcal{X})^{\otimes k} = \text{Span}\{U^{\otimes k} : U \in \mathcal{U}(\mathcal{X})\}. \quad (1.14)$$

Lemma 1.3.4 (Invariant subspace lemma). *Let $\mathcal{X}$ be a complex Euclidean space, let $\mathcal{V} \subseteq \mathcal{X}$ be a subspace, and let $A \in \mathcal{L}(\mathcal{X})$ be an operator. The following statements are equivalent:*

1. It holds that both $A\mathcal{V} \subseteq \mathcal{V}$ and $A^*\mathcal{V} \subseteq \mathcal{V}$.
2. It holds that $[A, \Pi_{\mathcal{V}}] = 0$.

Theorem 1.3.5 (Double Commutant Theorem). *Let $\mathcal{A}$ be a self-adjoint, unital subalgebra of $\mathcal{L}(\mathcal{X})$. Then the double commutant of $\mathcal{A}$ is equal to $\mathcal{A}$, i.e.,*

$$\text{Comm}(\text{Comm}(\mathcal{A})) = \mathcal{A}. \quad (1.15)$$

Proof. For all $X \in \text{Comm}(\mathcal{A})$ and any $A \in \mathcal{A}$, we have

$$[X, A] = AX - XA = 0. \quad (1.16)$$

Therefore, $A \in \text{Comm}(\text{Comm}(\mathcal{A}))$, so $\mathcal{A} \subseteq \text{Comm}(\text{Comm}(\mathcal{A}))$.

To demonstrate reverse inclusion, consider the subalgebra $\mathcal{B} \subseteq \mathcal{L}(\mathcal{X} \otimes \mathcal{X})$ defined as

$$\mathcal{B} = \{X \otimes \mathbb{I} : X \in \mathcal{A}\}. \quad (1.17)$$

For all $Y \in \mathcal{L}(\mathcal{X} \otimes \mathcal{X})$, it can be written in terms of the standard computational basis

$$Y = \sum_{i,j \in [d]} \sum_{a,b \in [d]} c_{i,j,a,b} |i\rangle\langle j| \otimes |a\rangle\langle b|. \quad (1.18)$$

Using the definition of the standard computational basis of operators, we can rearrange the right-hand side into the form

$$Y = \sum_{a,b \in [d]} Y_{a,b} \otimes E_{a,b} \quad (1.19)$$

where $Y_{a,b} = \sum_{i,j \in [d]} c_{i,j,a,b} |i\rangle\langle j|$.

Y commutes with $\mathcal{B}$ if and only if $Y(X \otimes \mathbb{I}) = (X \otimes \mathbb{I})Y$ for all $X \in \mathcal{A}$. Substituting Equation 1.19 and simplifying shows that Y commutes with $\mathcal{B}$ if and only if $[Y_{a,b}, X] = 0$. For any $X \in \text{Comm}(\text{Comm}(\mathcal{A}))$, X must commute with $\text{Comm}(\mathcal{A})$, and therefore

$$X \otimes \mathbb{I} \in \text{Comm}(\text{Comm}(\mathcal{B})). \quad (1.20)$$

It remains to show that every element in $\text{Comm}(\text{Comm}(\mathcal{B}))$ is of the form $X \otimes \mathbb{I}$. Define a subspace $\mathcal{V} \subseteq X \otimes X$, where

$$\mathcal{V} = \left\{ \text{vec}(X) = \sum_{a,b \in [d]} c_{a,b} |a\rangle \otimes |b\rangle : X \in \mathcal{A} \right\}. \quad (1.21)$$

For any arbitrary $X \in \mathcal{A}$, we find that

$$\begin{aligned} (X \otimes \mathbb{I})\mathcal{V} &= \{(X \otimes \mathbb{I})V : V \in \mathcal{V}\} \\ &= \left\{ (X \otimes \mathbb{I}) \left(\sum_{a,b \in [d]} c_{a,b} |a\rangle \otimes |b\rangle \right) : \sum_{a,b \in [d]} c_{a,b} |a\rangle \langle b| \in \mathcal{A} \right\} \\ &= \left\{ \sum_{a,b \in [d]} c_{a,b} X|a\rangle \otimes |b\rangle : \sum_{a,b \in [d]} c_{a,b} |a\rangle \langle b| \in \mathcal{A} \right\} \\ &\subseteq \mathcal{V}, \end{aligned} \quad (1.22)$$

the inclusion of which follows from the fact that $\mathcal{A}$ is an algebra so $X|a\rangle = Xe_a \in \mathcal{A}$. Similarly, since $\mathcal{A}$ is self-adjoint,

$$(X^* \otimes \mathbb{I})\mathcal{V} \subseteq \mathcal{V}. \quad (1.23)$$

By Lemma 1.3.4, this implies $[X \otimes \mathbb{I}, \Pi_{\mathcal{V}}] = 0$, and hence

$$\Pi_{\mathcal{V}} \in \text{Comm}(\mathcal{B}). \quad (1.24)$$

Therefore, for any arbitrary $X \in \text{Comm}(\text{Comm}(\mathcal{A}))$, Equation 1.20 implies

$$X \otimes \mathbb{I} \in \text{Comm}(\text{Comm}(\mathcal{B})). \quad (1.25)$$

It follows from Equation 1.24 that

$$[X \otimes \mathbb{I}, \Pi_{\mathcal{V}}] = 0. \quad (1.26)$$

By Lemma 1.3.4, this yields

$$(X \otimes \mathbb{I})\mathcal{V} \subseteq \mathcal{V}. \quad (1.27)$$

Since the subalgebra $\mathcal{A}$ is unital, $\mathbb{I} \in \mathcal{A}$ and thus $\text{vec}(\mathbb{I}) \in \mathcal{V}$. Therefore,

$$\begin{aligned} (X \otimes \mathbb{I})\text{vec}(\mathbb{I}) &= (X \otimes \mathbb{I}) \sum_{a \in [d]} |a\rangle \otimes |a\rangle \\ &= \sum_{a \in [d]} X|a\rangle \otimes |a\rangle = \sum_{a \in [d]} \left(\sum_{b,c \in [d]} X_{b,c} |b\rangle \langle c| \right) |a\rangle \otimes |a\rangle \\ &= \sum_{a \in [d]} \sum_{b \in [d]} X_{b,a} |b\rangle \otimes |a\rangle \\ &= \sum_{b,a \in [d]} X_{b,a} |b\rangle \otimes |a\rangle = \text{vec}(X) \in \mathcal{V}. \end{aligned} \quad (1.28)$$

By the bijectivity of vec , $X \in \mathcal{A}$, so $\text{Comm}(\text{Comm}(\mathcal{A})) \subseteq \mathcal{A}$. □

Theorem 1.3.6 (Schur-Weyl duality). *Let $\mathcal{X}$ be a complex Euclidean space, k be a positive integer, and $X \in \mathcal{L}(\mathcal{X}^{\otimes k})$ be an operator. Then the set of all commutators of the unitary group is equivalent to the span of the permutation operators, i.e.,*

$$\text{Comm}(\mathcal{U}(\mathcal{X}), k) = \text{Span}\{W_{\pi} : \pi \in S_k\} \quad (1.29)$$

Proof. Let $X \in \text{Comm}(\mathcal{U}(\mathcal{X}), k)$. then by the bilinearity of the Lie bracket,

$$X \in \text{Comm}(\text{Span}\{U^{\otimes k} : U \in \mathcal{U}(\mathcal{X})\}) \quad (1.30)$$

Proposition 1.3.2 implies that $X \in \text{Comm}(\mathcal{L}(\mathcal{X})^{\otimes k})$, so

$$\text{Comm}(\text{Span}\{U^{\otimes k} : U \in \mathcal{U}(\mathcal{X})\}) = \text{Comm}(\mathcal{L}(\mathcal{X})^{\otimes k}). \quad (1.31)$$

By Definition 1.2.1, $Z \in \mathcal{L}(\mathcal{X})^{\otimes k}$ if and only if there exists some $X \in \mathcal{L}(\mathcal{X})$ such that

$$Z = X = W_{\pi} X W_{\pi}^* \text{ for all } \pi \in S_k. \quad (1.32)$$

Right-multiplying and rearranging by W_{π} yields $XW_{\pi} - W_{\pi}X = 0$, and so by the bilinearity of the Lie bracket,

$$Z = X \in \text{Comm}(\text{Span}\{W_{\pi} : \pi \in S_k\}). \quad (1.33)$$

Hence

$$\mathcal{L}(\mathcal{X})^{\otimes k} = \text{Comm}(\text{Span}\{W_{\pi} : \pi \in S_k\}). \quad (1.34)$$

By Theorem 1.3.5, taking the commutant of both sides yields

$$\text{Comm}(\mathcal{L}(\mathcal{X})^{\otimes k}) = \text{Span}\{W_{\pi} : \pi \in S_k\}. \quad (1.35)$$

By substitution of Eq. 1.31, we have shown

$$\text{Comm}(\mathcal{U}(\mathcal{X}), k) = \text{Span}\{W_{\pi} : \pi \in S_k\}. \quad (1.36)$$

□